VI– Semester

Examination Scheme for EoSE-

CA – Continuous Assessment
EoSE – End of Semester Examination

<u>For Regular Students</u> –

Type of Examination	Course Code / Nomenclature	Duration of Examination		Maximum Marks		Minimum Marks	
Theory	CHM-76T-303. Bioinorganic chemistry, Organometallic chemistry, Heterocyclic chemistry, Carbohydrates, Spectroscopy, Quantum Mechanics and MOT.	CA	1 Hr.	CA	20	CA	8
		EoSE	3 Hrs.	EoSE	80	EoSE	32
Practical	CHM-76P-304 – Practical VI	CA	1 Hr.	CA	10	CA	4
		EoSE	4 Hrs.	EoSE	40	EoSE	16

The question paper in EoSE (End of Semester Examination) will consist of two parts A & B.

PART – A: 20 Marks

Part A will be compulsory, consisting of 10 very short answer-type questions (with a limit of 20 words) covering the entire syllabus, carrying two marks for each.

PART – B: 60 Marks

Part B of the question paper will have four questions with internal choice comprising of question number 2-5 which will be divided into four units. There will be one question with internal choice from each unit. Each question will carry 15 marks.

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For Non-Collegiate Students –

Type of Examination	Course Code and Nomenclature	Duration of Examination	Maximum Marks	Minimum Marks
Theory	**CHM-76T-303 –Bioinorganic chemistry, Organometallic chemistry, Heterocyclic chemistry, Carbohydrates, Spectroscopy, Quantum Mechanics and MOT**	**3 Hrs.**	**100**	**40**
Practical	**CHM-76P-304 Practical VI**	**4 Hrs.**	**50**	**20**

The question paper in EoSE (End of Semester Examination) will consist of two parts A & B

PART – A: 20 Marks

Part A will be compulsory, consisting of 10 very short answer-type questions (with a limit of 20 words) covering the entire syllabus, carrying two marks for each.

PART – A: 80 Marks

Part B of the question paper will have four questions with internal choice comprising of question number 2-5 which will be divided into four units. There will be one question with internal choice from each unit. Each question will carry 20 marks.

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SYLLABUS

[UG0802/03]-[CHM-76T-303]- Bioinorganic chemistry, Organometallic chemistry Heterocyclic chemistry, Carbohydrates, Spectroscopy, Quantum Mechanics and MOT

[UG0802/03]-[CHM-76P-304]- PRACTICAL-VI

VI – Semester – [Chemistry]

<table>
<tr><th>Semester</th><th>Code of the Course</th><th colspan="4">Title of the Course/Paper</th><th>NHEQF Level</th><th>Credits</th></tr>
<tr><td>VI</td><td>CHM-76T-303</td><td colspan="4">Bioinorganic chemistry, Organometallic chemistry Heterocyclic chemistry, Carbohydrates, Spectroscopy, Quantum Mechanics and MOT</td><td>5</td><td>4</td></tr>
<tr><td>VI</td><td>CHM-76P-304</td><td colspan="4">PRACTICAL-VI</td><td>5</td><td>2</td></tr>
<tr><td rowspan="2">Level of Course</td><td rowspan="2">Type of the Course</td><td colspan="3">Credit Distribution</td><td rowspan="2">Offered to NC Students</td><td colspan="2" rowspan="2">Course Delivery Method</td></tr>
<tr><td>Theory</td><td>Practical</td><td>Total</td></tr>
<tr><td>7</td><td>Major</td><td>4</td><td>2</td><td>6</td><td>Yes</td><td>Through Lecture, Sixty (60) Lectures</td><td>Class room Teaching/Power-Point (PPT)</td></tr>
<tr><td colspan="2">List of Programme Codes in which offered as Minor Discipline</td><td colspan="6">-NA-</td></tr>
<tr><td colspan="2">Prerequisites/Eligibility</td><td colspan="6">Every student automatically promoted from V to VI semester after the V-Semester EoSE.</td></tr>
<tr><td colspan="2">Course Objectives:</td><td colspan="6">The main objective of this course is to provide students with a theoretical understanding of advanced concepts in chemistry emphasizing both fundamental principles and practical applications.
Bioinorganic chemistry is introduced to study the role of metals in biological systems and their coordination environments.
Organometallic chemistry is explored including the structure, bonding and reactivity of metal-carbon bonds along with the</td></tr>
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	chemistry of inorganic polymers with their synthesis, properties and applications. The syntheses and reactivity of heterocyclic compounds and carbohydrates have been incorporated to achieve knowledge and deep understanding in these fields. Spectroscopic techniques are incorporated, viz. rotational and vibrational spectroscopy to understand molecular rotations and vibrations, raman spectroscopy for complementary vibrational analyses, electronic spectroscopy for studying electronic transitions in molecules. Principles of quantum mechanics and its applications in molecular orbital theory (MOT) are also introduced for understanding molecular structure and bonding. This course integrates theoretical concepts of spectroscopy and quantum mechanics to prepare students for advanced research and problem-solving in modern chemistry.

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Detailed Syllabus

VI– Semester – [Chemistry]

CHM-76T-303- **(4 Hrs./week)** **Bioinorganic chemistry, Organometallic chemistry, Heterocyclic chemistry, Carbohydrates, Spectroscopy, Quantum Mechanics and MOT.**

Unit-I

Bioinorganic chemistry:
Essential and trace elements to biological processes, Metalloporphyrin with special reference to hemoglobin and myoglobin. Biological role of alkali and alkaline earth metal ions with special reference to Ca^{2+}, Nitrogen fixation.

Organometallic Compounds:
Definition, nomenclature and classification of organometallic compounds. Preparation, properties, bonding and applications of alkyls and aryls of Li, Al, Hg, Sn and Ti, A brief account of metal ethylenic complexes and homogeneous hydrogenation, Mononuclear carbonyls and the nature of bonding in metal carbonyls.

Inorganic Polymers:
Silicones and phosphazenes as examples of inorganic polymers, nature of bonding in triphosphazenes.

15 Lecture

Unit-II

Heterocyclic Compounds
Introduction: Molecular orbital diagram and aromatic characteristics of pyrrole, furan, thiophene and pyridine. Methods of synthesis and chemical reactions with particular emphasis on the mechanism of electrophilic substitution. Mechanism of nucleophilic substitution reactions in pyridine and derivatives. Comparison of basicity of pyridine, piperidine and pyrrole.

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Introduction to condensed five- and six-membered heterocyles. Preparation and reactions of indole, quinoline and isoquinoline with special reference to Fisher-indole synthesis, Skraup synthesis and Bischler-Napieralski synthesis, Mechanism of electrophilic substitution reactions of indole, quinoline and isoquinoline.

Carbohydrates: Introduction, classification, constitution and reaction of glucose and fructose, Mutarotation and its mechanism, Cyclic structure : pyranose and furanose forms, Haworth projection formulae, Configuration of monosaccharides, Determination of ring size, Conformational analysis of monosaccharides, Epimerization, Chain lengthening and chain shortening in aldoses. Interconversion of aldoses and ketoses.
Disaccharides: Structure determinations of maltose, lactose and sucrose.
Polysaccharides: Structure of starch and cellulose, glycosidic linkages.

15 Lecture

Unit-III

Spectroscopy:

Introduction: Electromagnetic radiation, Spectrum, Basic features of different spectrometers, Statement of the Born-Openheimer approximation, Degrees of freedom.
Rotational Spectrum: Diatomic molecules, Energy levels of a rigid rotator (semi-classical principles), Selection rules, Spectral intensity using population distribution (Maxwell-Boltzmann distribution), Determination of bond length, Qualitative description of non-rigid rotator, Isotope effect.
Vibrational Spectrum: Infrared spectrum: Energy levels of simple harmonic oscillator, Selection rules, Pure vibrational spectrum, intensity, determination of force constant and qualitative relation of force constant and bond energies, effect of anharmonic motion and isotope on the spectrum, Idea of vibrational frequencies of different functional groups.
Raman Spectrum: Basic principles and applications, Concept of polarizability, Pure rotational and pure vibrational Raman spectra of diatomic molecules, Selection rules
Electronic Spectrum: Concept of potential energy curves for bonding and antibonding molecular orbitals, Qualitative description of selection rules and Frank Condon principle. Qualitative description of σ, π and n M.O. and their respective energy levels.

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15 Lecture

Unit-IV

Elementary Quantum Mechanics:
Black-body radiation, Planck's radiation law, Photoelectric effect, Heat capacity of solids, Bohr's mode of hydrogen atom (no derivation) and its defects. Compton effect.
De Broglie hypothesis, The Heisenberg's uncertainty principle, Sinusoidal wave equation, Hamiltonian operator, Schrodinger wave equation and its importance, Physical interpretation of the wave function, Postulates of quantum mechanics, quantum number and their importance, Particle in a one-dimensional box.

Molecular orbital theory:

Basic ideas, criteria for forming M.O. from A.O. Construction of M.O's by LCAO method- H_2^+ ion, calculation of energy level from wave functions, physical picture of bonding and antibonding wave functions, concept of σ, σ^*, π, π^* orbitals and their characteristics. Hybrid orbitals - sp, sp^2, sp^3, calculation of coefficients of A.O.'s used in these hybrid orbitals. Introduction to valence bond model of H_2, comparison of M.O. and V.B. models.

Suggested Books and References:

1. Bioorganic, bioinorganic and supramolecular chemistry by Kalsi and Ashu , New Age International.
2. Bioinorganic Chemistry by Ankita Das, Asim K Das and Mahua Das. Books & Allied.
3. Principal of bioinorganic Chemistry by Stphen J. Lippard, Jerem M. Berg, Universal Science Books.
4. Basic Organometallic Chemistry by B D Gupta, A. J. Elias, Universities Press
5. Organometallic Chemistry and Catalysis by Didier Astruc, Springer.
6. Organic Chemistry by R. T. Morrison and R. N. Boyed, Prentice Hall.
7. Organic Chemistry by I. L. Finar (Vol. I & II), ELBS.
8. Advanced Organic Chemistry by A. Bahl and B**. S.** Bahl, S. Chand.
9. Organic Chemistry by S. S. Gupta, Oxford University Press.
10. Modern Organic Chemistry by M.K. Jain and S. C. Sharma, Vishal Publishing Co.
11. Essentials of carbohydrate Chemistry by John F. Robyt, Springer
12. Heterocyclic Chemistry By R.R Gupta, Springer.
13. Heterocyclic Chemistry by R K Bansal New Age International Publishers.
14. Principles of Physical Chemistry by B. R. Puri, L. R. Sharma and M. S. Pathania, Vishal Publishing Co.
15. Advanced Physical Chemistry by Gurdeep Raj, Goel Publishing House.
16. Atkins' Physical Chemistry by Atkins, Julio De Paula and James Keeler, Oxford.
17. Quantum Chemistry (2nd edition) by Donald A. McQuarrie, University Science Book Sausalito, California.

Suggested E-resources:

All the above suggested books are **available as e- books**.

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Online Lecture Notes and Course Materials:

All prescribed courses are available in digital form in the form of e-books, Adobe Acrobat documents (PDF), web pages etc.

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Syllabus

CHM-76P-304: Practical VI 4 Hrs./week

Inorganic chemistry **10 marks**

Quantitative Analysis:

a. Determination of Dissolved Oxygen in water.
b. Determination of Chemical Oxygen Demand (COD)
c. Determination of Biological Oxygen Demand (BOD)
d. Percentage of available chlorine in bleaching powder.
e. Estimation of total alkalinity of water samples (CO_3^{2-}, HCO_3^-) using double titration method.
f. Measurement of chloride, sulphate and salinity of water samples by simple titration method ($AgNO_3$ and potassium chromate).
g. Separation and estimation of Mg (II) and Fe (II) by solvent extraction method.
h. Separation and estimation of Mg (II) and Fe (II) by ion exchange method.

Organic Chemistry **10 marks**

Synthesis of Organic Compounds

a. **Acetylation:** acetylation of salicylic acid, aniline, glucose and hydroquinone.
b. **Benzoylation:** benzoylation of aniline and phenol.
c. **Aliphatic electrophilic substitution:** preparation of iodoform from ethanol or acetone

d. **Nucleophilic addition elimination:** preparation of semicarbazone of acetone, ethyl methyl ketone, cyclohexanone and benzaldehyde.

e. **Aromatic electrophilic substitution:**
   - **Nitration:** Preparation of m-dinitrobenzene
   - Preparation of p-nitroacetanilide
   - **Halogenation:** Preparation of $p - bromoacetanilide$
   - Preparation of 2, 4, 6 – tribromophenol
   - **Diazotization/coupling:** Preparation of methyl orange or methyl red

f. **Oxidation:** Preparation of benzoic acid from toluene or benzaldehyde

g. **Reduction:**
   - Preparation of aniline from nitrobenzene
   - Preparation of m-nitroaniline from m-dinitrobenzene

Physical Chemistry **10 marks**

1. **pH-metry**
   - **a. pH metric titrations of**
     - i. Strong acid vs. strong base
     - ii. Weak acid vs. strong base
   - **b.** Determination of dissociation constant of a weak acid.
   - **c. Preparation of buffer solutions:**
     - i. Sodium acetate-acetic acid
     - ii. Ammonium chloride-ammonium hydroxide
   - **d.** Measurement of the pH of buffer solutions and comparison of the values with theoretical values.

2. **Potentiometry**
   - **a.** Strong acid vs. strong base
   - **b.** Weak acid vs. strong base
   - **c.** Potassium dichromate vs. Mohr's salt

3. **Spectrophotometery or Colourimetry:**
   To verify Beer-Lambert law and determine the concentration of the given aqueous solution of $KMnO_4/K_2Cr_2O_7/CuSO_4$ of unknown concentration.

Viva voce **5 marks**

Practical Record **5 marks**

Suggested Books and References:

1. Vogel's Qualitative Inorganic Analysis, A. I. Vogel Prentice Hall.
2. Vogel's Quantitative Inorganic Analysis Including Elementary Instrumental Analysis, ELBS.

3. Vogel's Textbook of Quantitative Chemical Analysis, A. I. Vogel, Pearson Education Ltd.
4. Advanced Practical Organic Chemistry by N. K. Vishnoi, Vikas Publishing House Pvt Ltd.
5. Comprehensive Practical Organic Chemistry: Preparation and Quantitative Analysis, V. K Ahluwalia. Universities Press, Hyderabad.
6. Laboratory Techniques in Organic Chemistry by V. K Ahluwalia, I K International, N
7. Advanced Practical Organic Chemistry J. B Yadav, Goel Publishing House.
8. Practical Physical Chemistry, by B. D Khosla, S. Chand & Company.
9. Advanced Practical Organic Chemistry by Amit Arora, Discovery Publishing House, New Delhi.

Suggested E-resources:

All the above suggested books are available as e- books.

Online Lecture Notes and Course Materials:

All prescribed courses are available in digital form in the form of e-books, Adobe Acrobat documents (PDF), web pages etc.

Course Learning Outcomes:

By the end of this course, students will get a clear understanding of bioinorganic chemistry including the role of metals in biological systems and their coordination environments. They will get acquainted with the structure, bonding, reactivity and the applications of organometallic complexes along with the synthesis, properties, and applications of heterocyclic compounds. Furthermore, the structure, functions and reactions of carbohydrates will provide knowledge in this field. Students will get knowledge in spectroscopic techniques through analysis of molecular rotation using the rotational spectrum, examine molecular vibrations through the vibrational spectrum and Raman spectrum and further interpret electronic spectra for electronic transitions in molecules. They will grasp the fundamentals of quantum mechanics and its application in quantum molecular orbital theory (MOT) to explain molecular bonding and structure.

This course provides students with both theoretical knowledge and practical skills, essential for advanced studies and research in modern chemistry.

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